LEWICE 2.0: Background and Validation

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Outline

- **Background**
- **LEWICE 2.0 Features**
- **Software Management Process**
- **Validation Tests**
- **Quantitative Method**
- **Quantitative Comparison**
  - **Experimental Variation**
  - **LEWICE Variation**
- **Comparison to Qualitative Approach**
- **Statistical Variations**
- **Conclusions**

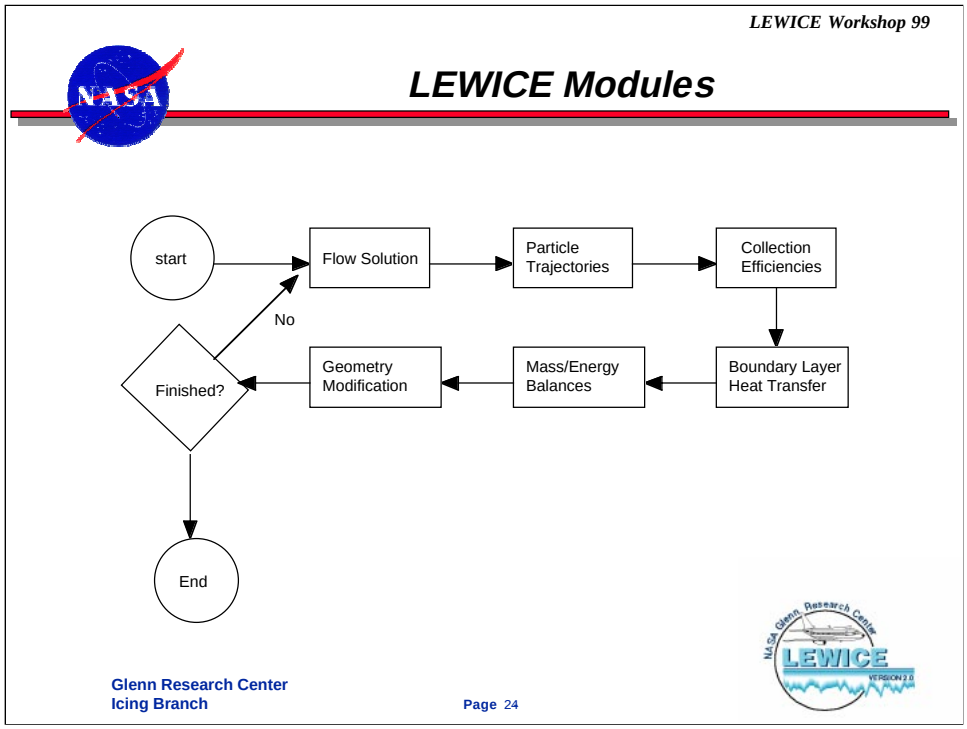
This presentation will outline the features of LEWICE 2.0. It will also cover our validation process which includes software management, version control, validation tests and the technique used to compare LEWICE 2.0 to the experimental data.

Background

- **Development initiated in early 1980's**
- **First general release: 1991 (Version 1.0)**
- **Major Updates:**
  - **Version 1.3 (1993)**
  - **Version 1.6 (1995)**
  - **Version 2.0 (1999)**

The NASA Glenn Icing Branch has developed several computer codes which are designed for use by industry as aids for certification into known icing conditions and for the design of ice protection systems.

LEWICE stands for LEWis ICE accretion codes and is loosely defined as representing any of the codes developed by or for the NASA icing branch. There are no plans at present to rename the codes to reflect the center's name change.



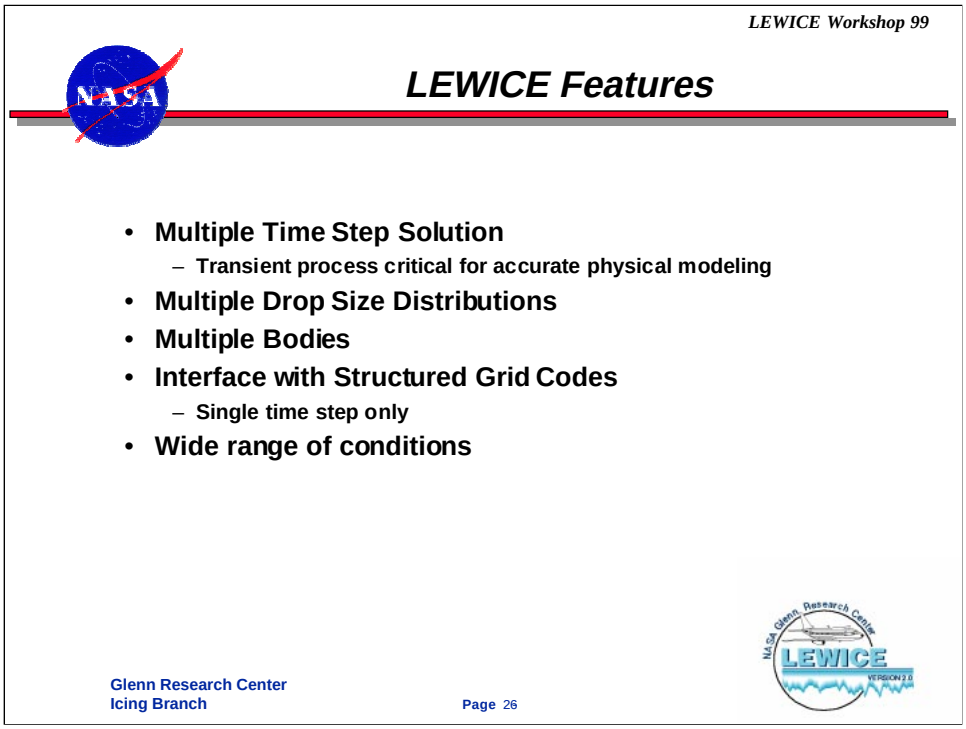
This diagram graphically shows the LEWICE process. After reading in the user inputs, it calculates a flow solution which is then used to track individual water particles to the airfoil surface. From these trajectories, impingement limits and collection efficiencies are found. The outer surface heat transfer coefficient is found from a boundary layer analysis. All of this data is then used to calculate a predicted ice thickness at the surface using standard mass and energy transfer equations. The ice thickness is used to deform the surface to grow the ice. These steps are repeated until the final icing time has been reached.

LEWICE Modules

- **Flow Solution**
  - Panel (default)
  - Naviér-Stokes
    - » Reads in solution
    - » Single time step only
- **Particle Trajectories**
  - Collection efficiency
  - Impingement limits
- **Integral Boundary Layer**
  - Heat transfer coefficient
- **Mass/Energy Balance**
- **Geometry Modification**

The baseline code consists of a 2D potential flow code, an integral boundary layer code, a particle trajectory code, a module which solves for the simultaneous heat and mass transport on the surface, and a routine to add ice growth to the surface.

An accurate method for predicting ice shapes relies on being able to track changes to the system as the ice grows. Hence it is necessary to pass through the above loop several times for typical accretions. A critical step in the process is to define, in an automated fashion, a new geometry for each time step. This geometry must accurately show the deflection of the surface due to ice growth while remaining smoothly contoured for computational purposes.

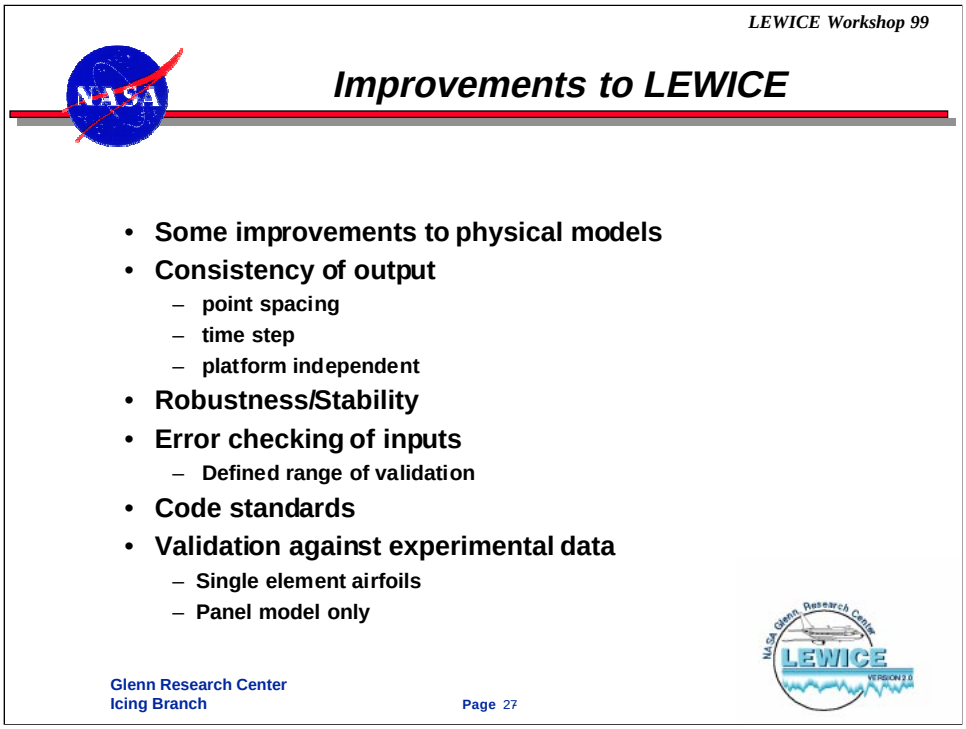


Users should run several time steps in a typical ice accretion to generate an accurate shape. The manual provides details on recommended settings. Typical time steps range from 30 seconds to 3 minutes.

Drop size distributions with up to 10 different drop sizes can be used. However, unless the data has a very skewed distribution or the user is simulating a multi-element airfoil, a single drop size distribution is usually sufficient.

The number of bodies is currently limited to 5 to reduce memory requirements. Wing/tail interactions and multi-element airfoils are two uses for this feature. Experimental data is very limited however, so the code cannot be validated for these configurations.

The interface with structured grid codes is limited to a single time step and is not useful for ice accretion. That portion of the model has not been thoroughly tested or validated. It may have use in calculating collection efficiencies on multi-element airfoil. Results would have to be compared to experimental data to validate the code for that purpose.

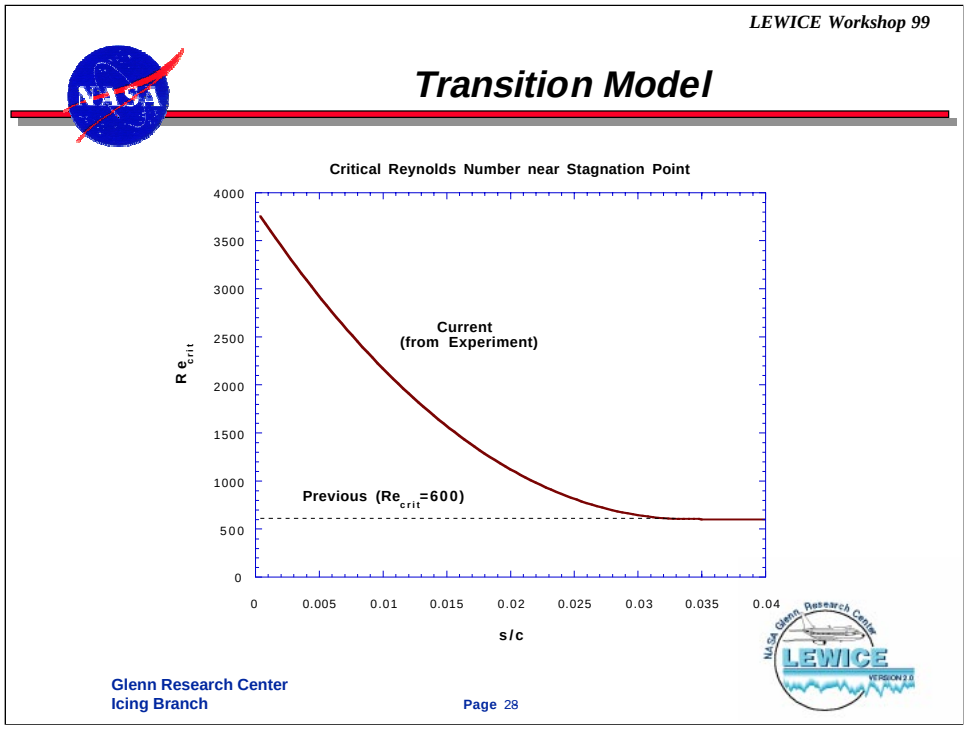


The method for predicting transition to turbulent flow has been changed to more accurately reflect experimental measurements of transition on ice shapes. The prrediction of sand-grain roughness was also modified based on experimental data.

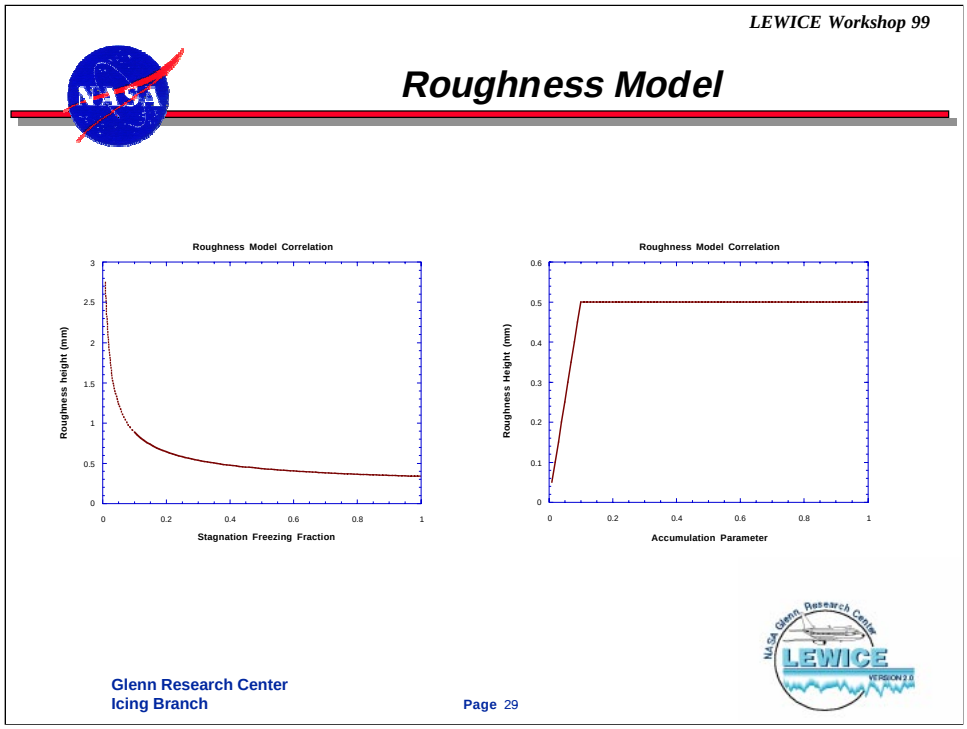
The reworking of several modules has resulted in more uniform output with changes in time step and point spacing. The execution speed is also at least 20% faster than version 1.6 which was released at the last workshop.

Hundreds of code runs were made during the validation and testing phases, many with the expressed intent of trying to crash the code. In one case, the code exceeded the array limits and stopped. All other cases ran to completion.

The program checks for errors in the input file and geometry file and informs the user when they are running a case which is outside of the validation runs. This does not mean that LEWICE will not run the case nor that the results are necessarily inaccurate, but that data is not available in that regime. The code has been validated against a wide range of experimental data.



Standard boundary layer theory states that transition starts when the Reynolds Number based on roughness height exceeds 600. However, that data did not look at roughness near the leading edge. Dr. Bragg and his students at the University of Illinois performed experiments near the leading edge to determine the transition point. The results above used hemispherical beads on a NACA0012 airfoil.

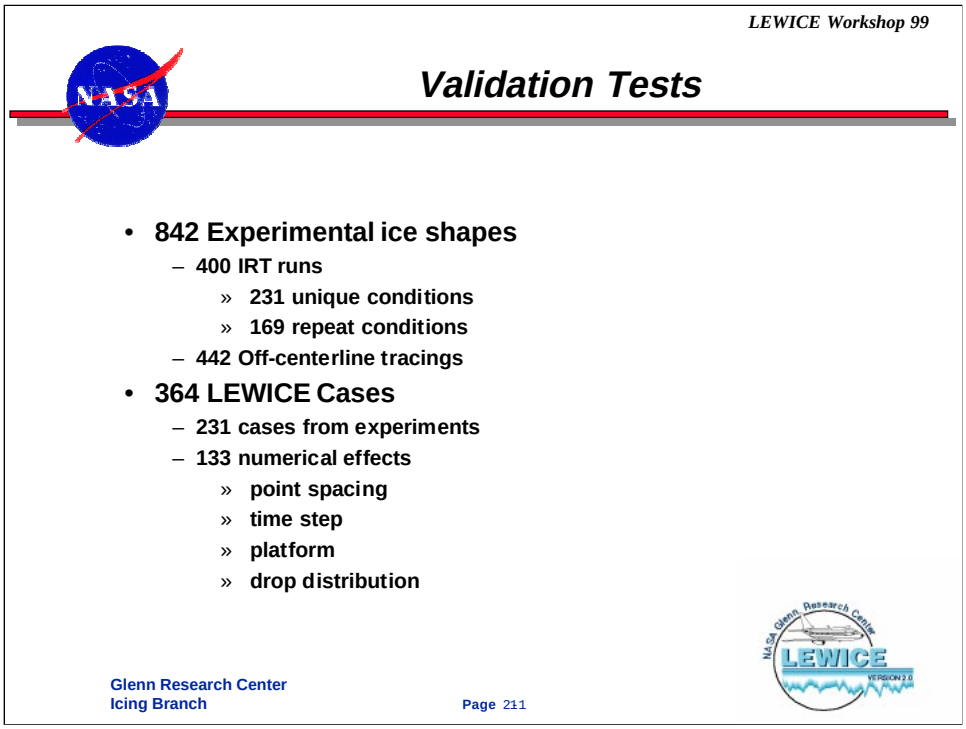


Ice roughness heights were measured for several conditions, first by Dr. Jai Shin in 1992 and later by Dr. Dave Anderson in 1997 and 1998. The correlation above takes into account the transient build up of roughness (the curve on the right) as well as the increasing roughness as conditions progress further into glaze ice conditions. The curve is extrapolated at very low freezing fractions.

Software Management

- **Document changes**
- **Revision tracking**
- **Standardized routines**
- **Detailed variable definitions**
- **More thorough subroutine testing**

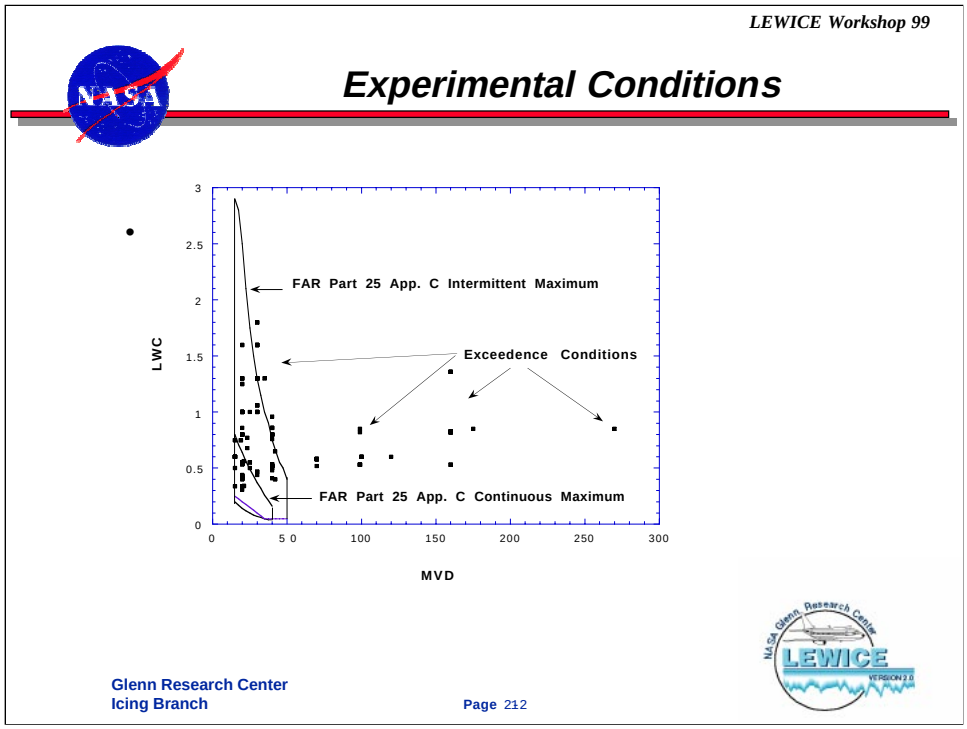
The software management process has improved our ability to find errors in the code and correct then faster. Better testing procedures results in a more robust, stable code. Improved version tracking will provide better support to users and better documentation of code results and code features. Structural changes in various subroutines allows for improved modularity.



The cases performed with this code comprise the most comprehensive set of validation runs ever attempted. The data set consists of every digitized ice shape available for publication on single element airfoils and covers data taken from 1991 to early 1998. In addition, several cases were run to determine code sensitivity to numerical factors such as point spacing, time step, computer platform and compilers.

This validation effort will likely be performed for any future releases of LEWICE so that quantitative comparisons can be made as to increases in accuracy.

The validation database (CR 208690) has been made publicly available so that others could make an independent assessment of the code's quality. Other researchers who are interested in validating other ice accretion codes can also use this database to validate their code. It is our opinion that other codes (either ours or others) will have to demonstrate higher quantitative accuracy on this database to be considered superior to LEWICE 2.0.



This plot shows the meteorological conditions used for this validation exercise. Most of the data lies within either the continuous maximum envelope or the intermittent maximum envelope, although there are some data points in exceedence conditions.

Experimental Data Range

- **Time: 2 to 45 minutes**
- **Chord: 14" to 78"**
- **AOA: -4° to 7°**
- **Velocity: 109 to 284 knots**
- **Reynolds Number: 2.3*106 to 1.3*107**
- **Mach Number: 0.17 to 0.45**
- **LWC: 0.31 to 1.8 g/m3**
- **MVD: 15 to 270 micron**
- **Static Temperature : -25 to 27 °F**
- **Total Temperature : -15 to 33 °F**

This table shows the range of the experimental data. Not all combinations of each extreme is available (see previous graph). LEWICE is capable of running conditions outside these ranges, but data is needed in order for the code to be validated for conditions outside those listed.

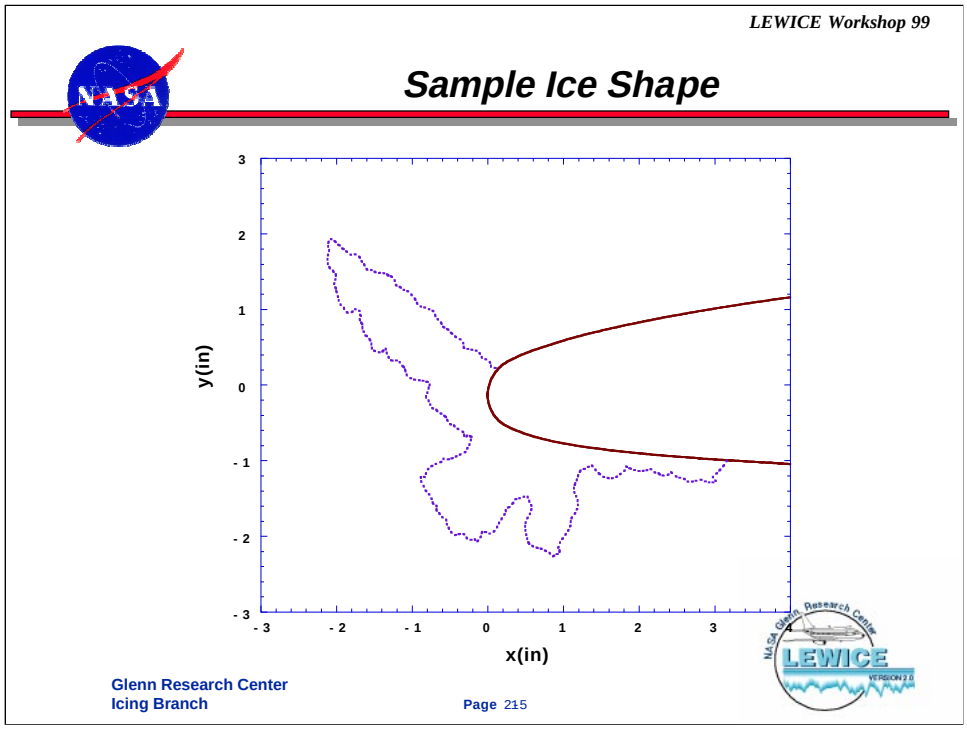
Ambient pressure and relative humidity are code inputs which are not normally recorded in the tunnel. Since the tunnel is not pressurized, a nominal value was selected near atmospheric pressure and the relative humidity was fixed at 100%. Both inputs are only secondary influences on the ice shape, hence small deviations from those input will not greatly affect the ice shape.

Ice Shape Comparison

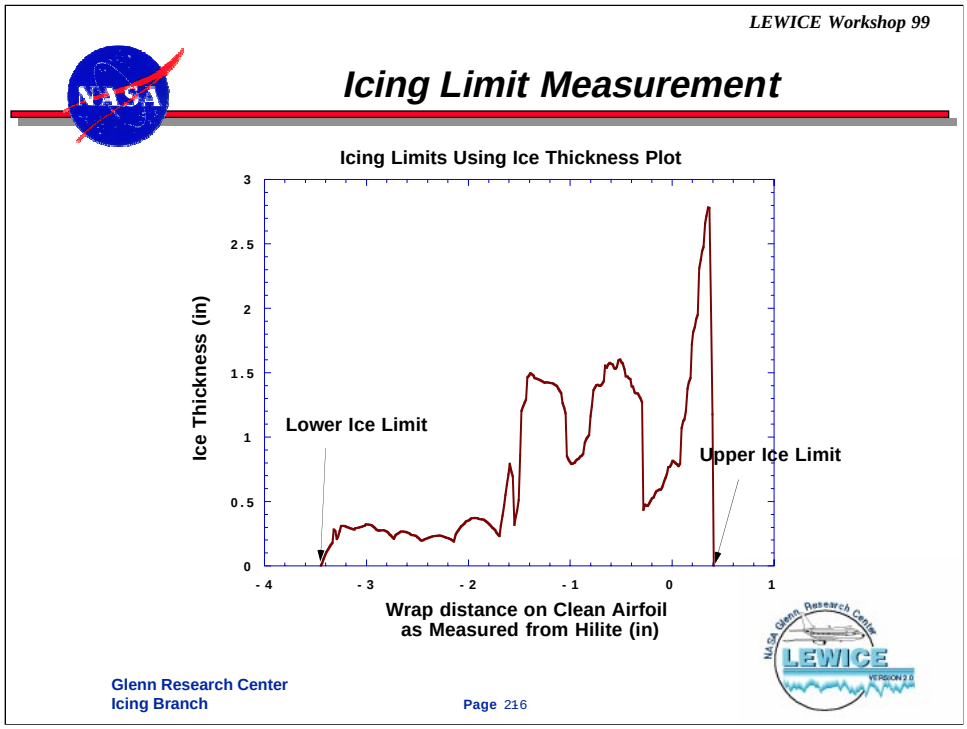
- **Quantitative measurement of important features**
  - **Icing Limits**
  - **Ice Thicknesses**
    - » **"Horn" Thickness**
    - » **"Stagnation" Thickness**
  - **Ice Area**
  - **Horn Angle**

In order to assess the accuracy of the results, quantitative measurements must be taken on both the experimental ice shape and the predicted ice shape. The features listed above were those considered in this validation approach.

Research continues into improving the shape comparison and automating the procedure.

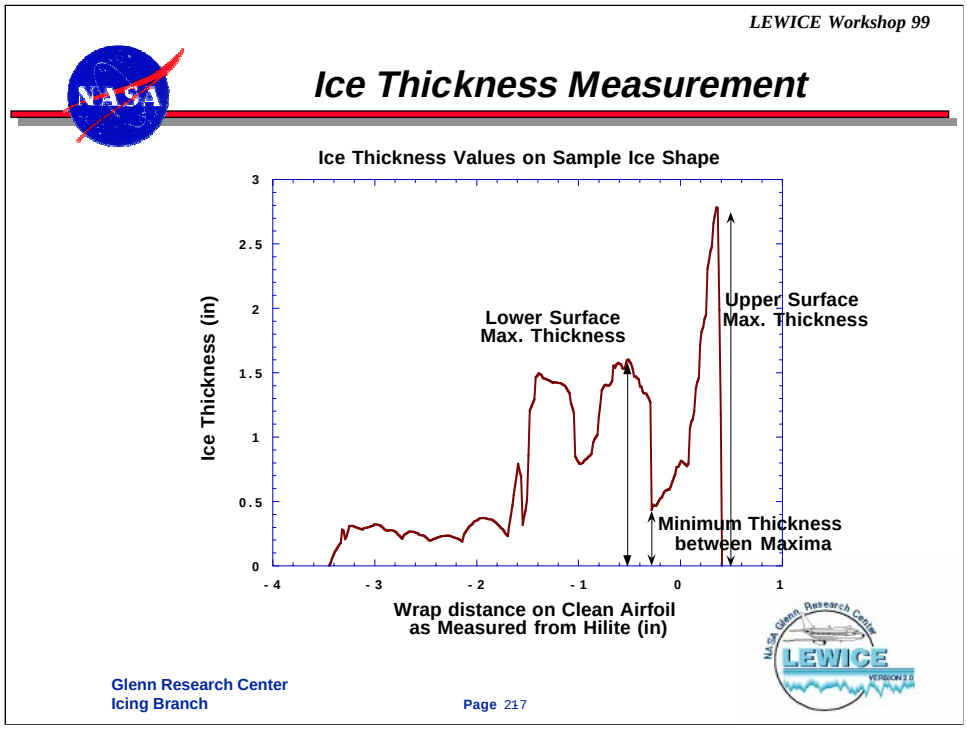


This 45 minute glaze ice shape will be used to demonstrate the measurement technique used.



This plot shows the ice thickness at each point on the airfoil as a function of the surface distance. The last location where the ice thickness goes to zero is considered the icing limit. This means that if there are ice feathers disconnected from the main ice shape, they are counted in the icing limit.

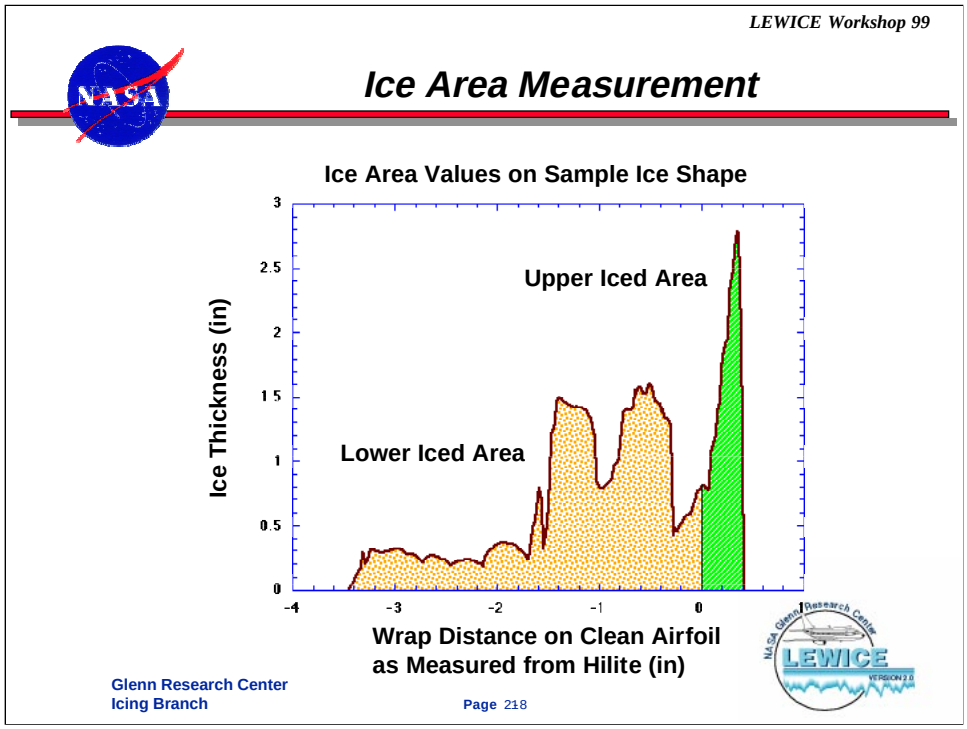
Hilite is used in this context as the minimum x-location on the airfoil.



The maximum ice thicknesses on the upper and lower surface define the location and height of the upper and lower ice horns. The minimum thickness between these two maxima is also recorded. In many cases, this is labeled “stagnation thickness” although the actual stagnation point may not reside at that exact location.

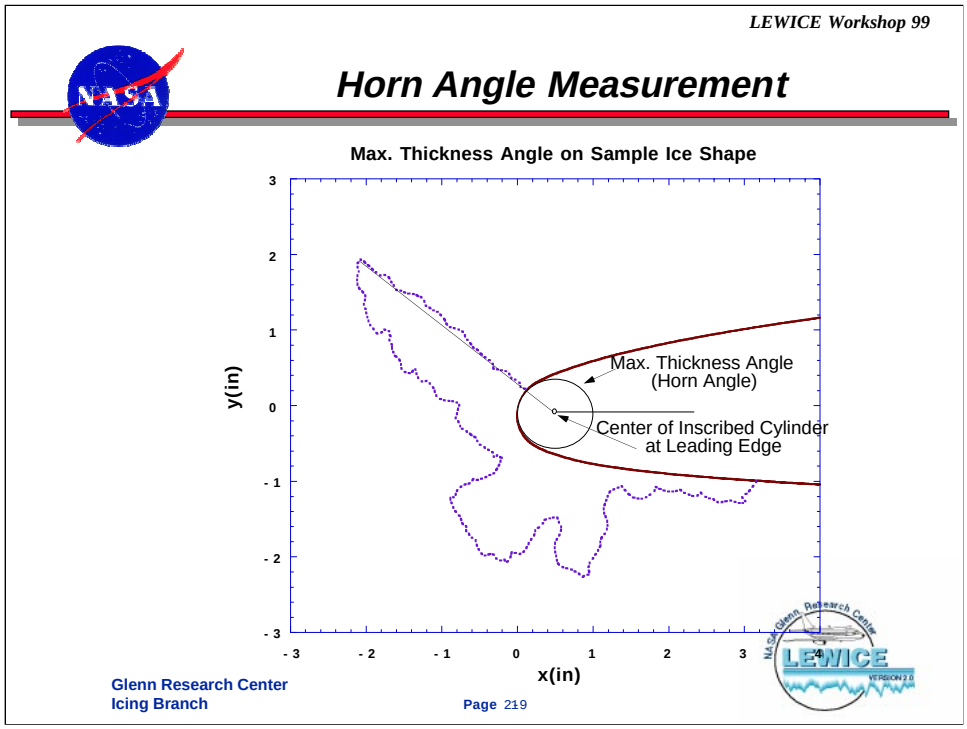
This figure also shows some of the difficulties in horn identification. There are two ice horns in the lower surface, each of which is approximately the same size. A very small difference in their respective heights could change which horn is selected, hence changing the horn angle.

Hilite is used in this context as the minimum x-location on the airfoil.



The ice area is measured two ways currently. In this figure, the thickness distribution is integrated to give ice area. This is the area given in the validation report. The area is also calculated directly by integrating the iced geometry coordinates and subtracting the area of the clean airfoil. The second method can possibly lead to truncation errors as the airfoil area is much larger than the iced area.

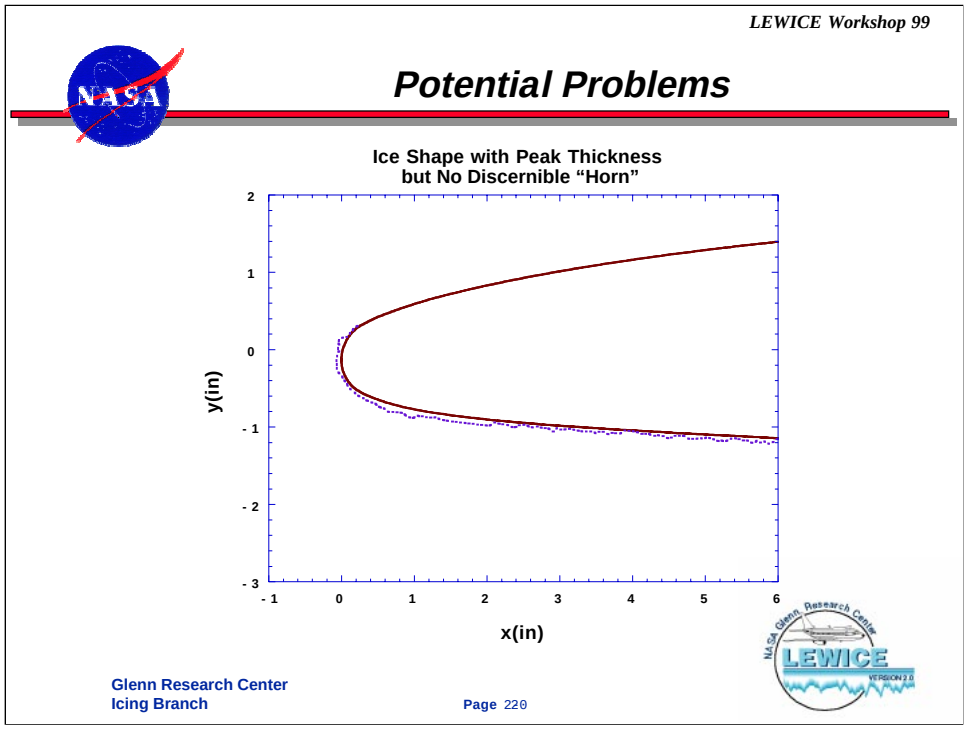
Hilite is used in this context as the minimum x-location on the airfoil.



For the validation report, the horn angle was measured using the center of the inscribed cylinder at the leading edge as a reference point.

Other techniques could use the leading edge of the airfoil or the surface normal as the reported angle.

Feedback is requested on which technique is most acceptable to the icing community.



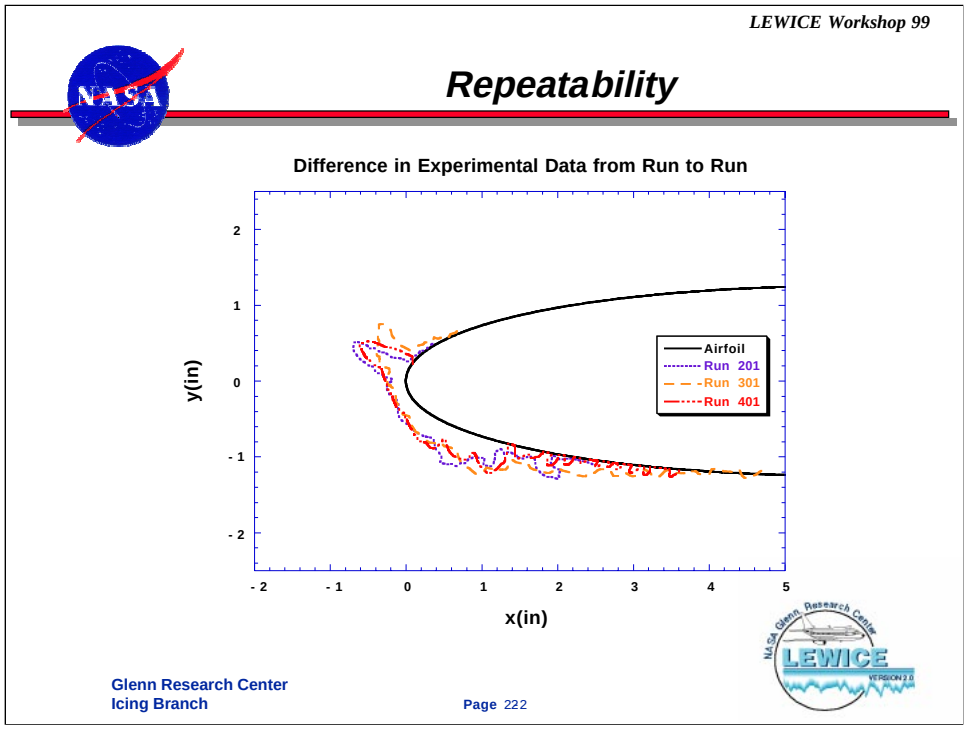
Small ice shapes without a discernable "horn" do not fit the measurement technique used. The branch is exploring alternative techniques to evaluate these shapes.

Quantitative Comparison

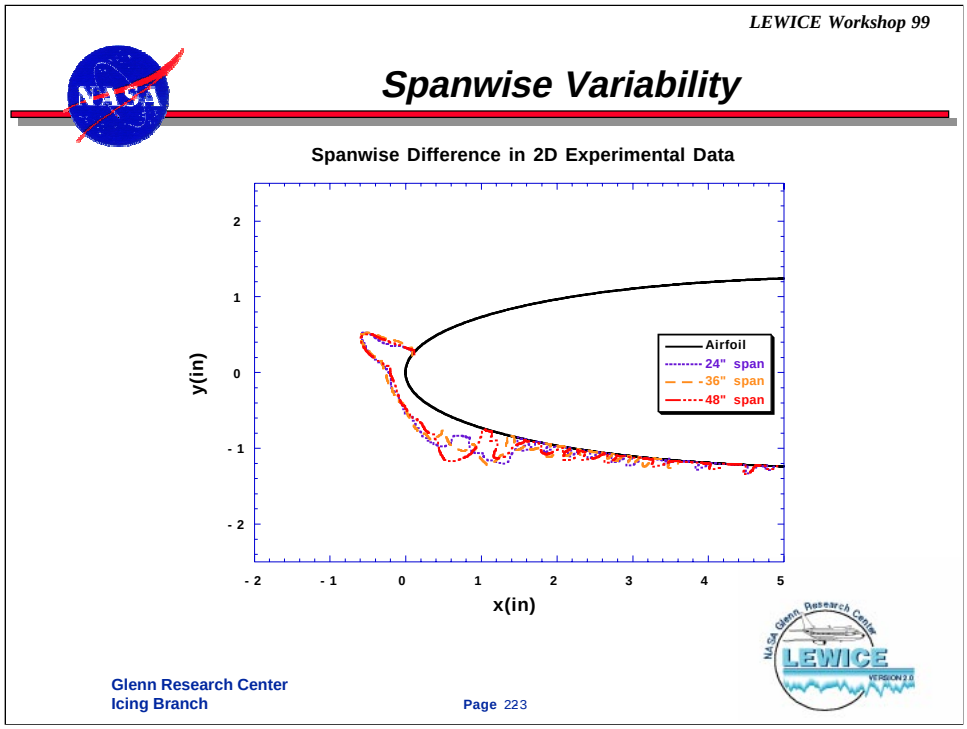
- **11 measurements taken on 1073 ice shapes (842 experimental and 231 LEWICE)**
- **Data entered into Excel® spreadsheet**
- **% differences calculated for each measurement**
- **Experimental data categorized by:**
  - **Repeatability**
  - **Spanwise Variability**
  - **Tracing technique**

A computer program was written to extract the 11 measurements from a given experimental or calculated ice shape. This utility program is supplied on the CD-ROM with the LEWICE 2.0 release. The program outputs a small data file containing the measurements. This data was then cut and pasted into the appropriate cells of the spreadsheet.

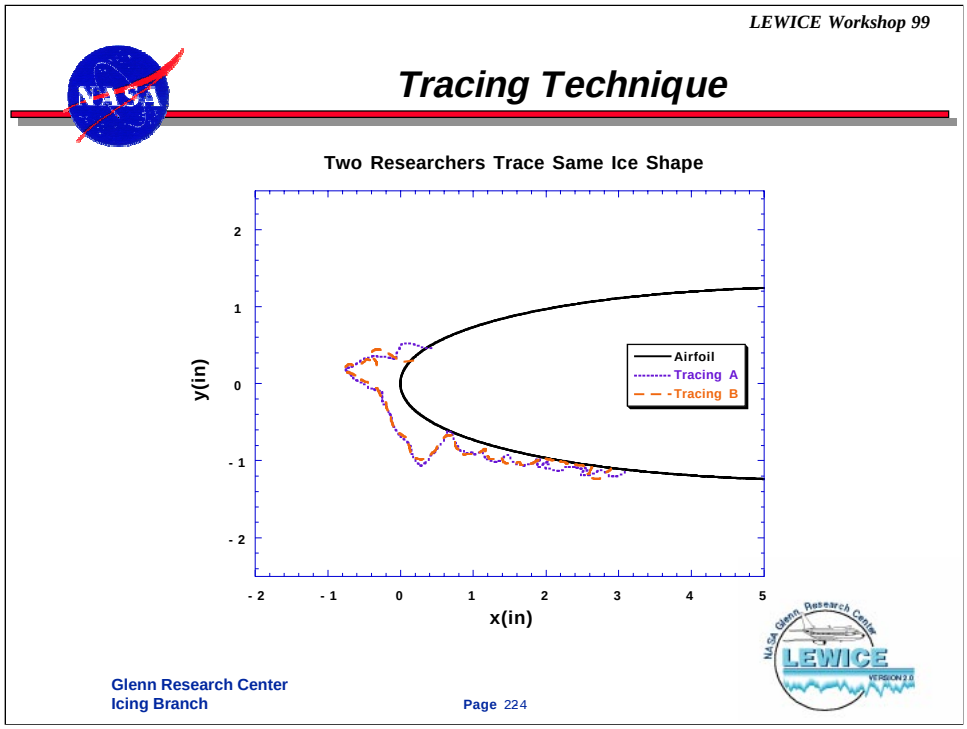
Spreadsheet calculations were then input to determine percent differences for each case.



This plot shows typical results from the assessment of tunnel repeatability. In this case, the same condition was run in the tunnel for this airfoil in 1996, 1997 and 1998. The quantitative repeatability of this condition was measured to be 2.6% which is the average value for repeatability of the IRT using our measurement technique.



This plot shows the typical variation along the span of the airfoil for a 2D model. Tracings were taken at 24”, 36”, and 48” from the floor of the test section. The quantitative repeatability of this condition was measured to be 2.5% which is the average value for spanwise variability of the IRT using our measurement technique. By coincidence, the condition shown corresponds to Run 401 from the previous slide.



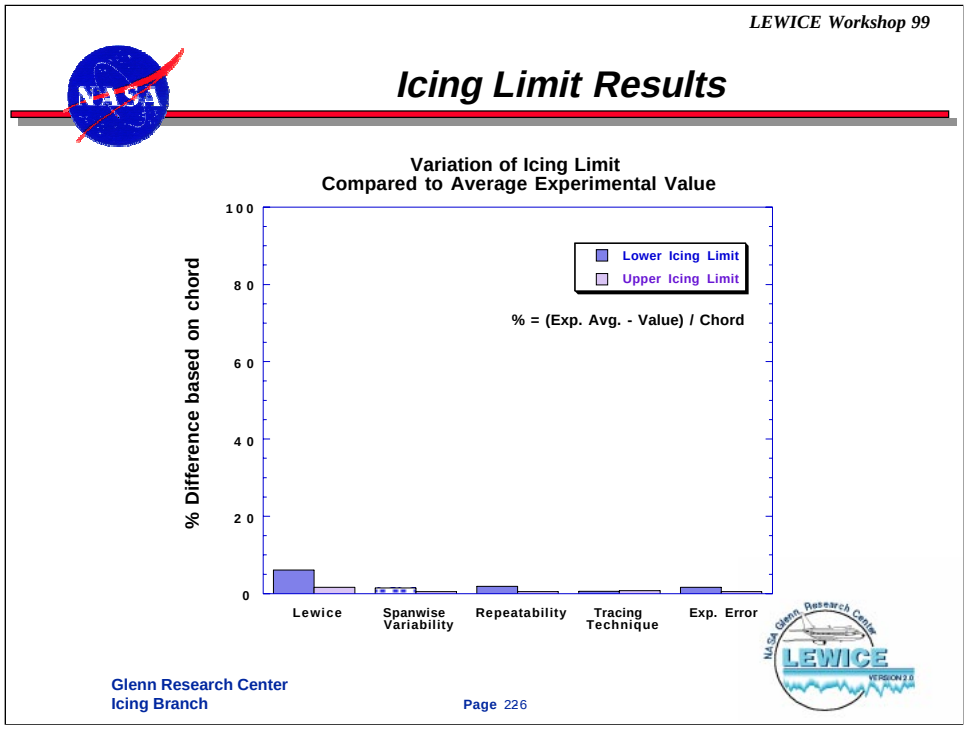
Ice shapes are recorded in the tunnel by first cutting out a section using a heated piece of metal. A cardboard template of the clean airfoil is inserted and the shape is traced with a pencil. The pencil tracing is then hand-digitized to provide the iced coordinates. Since all of these steps require human judgement, there could be discrepancies due to tracing technique. This plot shows a case where two researchers traced the same ice shape to asses this difference. The average variation due to this factor is 1.3%, or half the difference produced by repeatability or spanwise variability. It should also be noted that more than one ice tracing will only be taken when the researchers believe there is not uniformity in technique, so the actual error due to this process is likely much less.

Overall Comparison

- **Experimental data average consists of average of all individual tracings**
- **Overall comparison factor is average of individual % differences for each measurement**
- **Comparison involves non-dimensional values to normalize data**

Each of the measurements are tabulated in the spreadsheet and non-dimensionalized to eliminate differences in scale. The difference between two experimental ice shapes or an experimental ice shape and the LEWICE prediction are calculated in the spreadsheet. The average of the absolute differences are then plotted in the following bar charts. Since these are considered percent differences, the y-axis is fixed from 0% to 100%

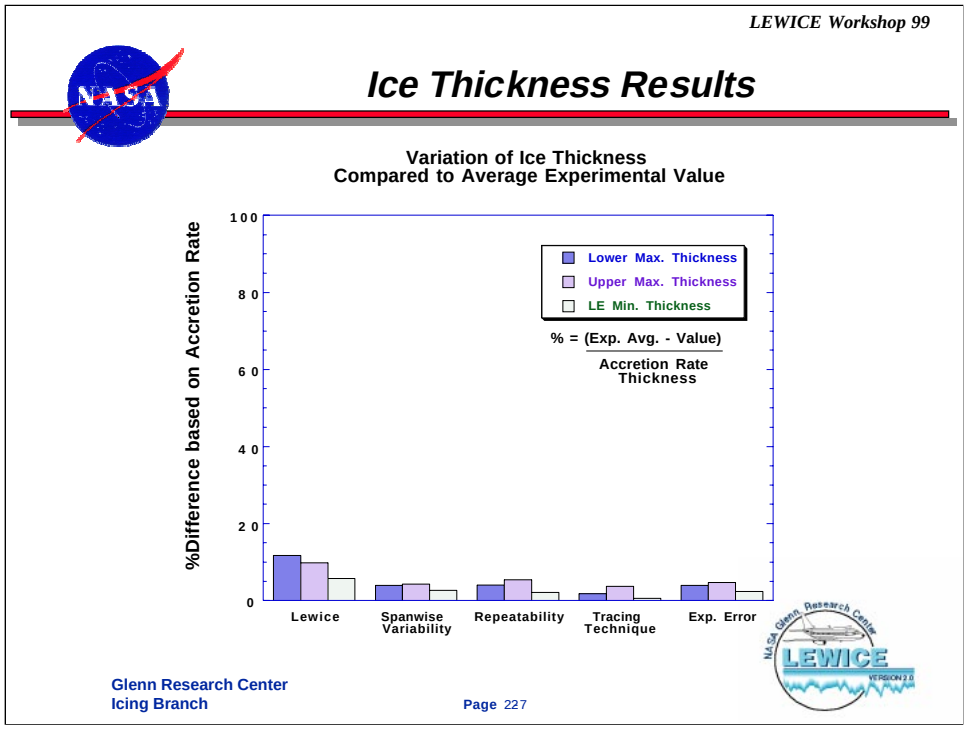
It should be noted that although the non-dimensionalized results are presented here, the Excel spreadsheet contains the actual measurements in inches, and the straight percent differences for these cases. The non-dimensionalized results are shown because they most closely agree with the qualitative assessment of the differences in ice shape. Research is continuing into other methods for quantitative comparison of the data sets.



This plot shows the average differences for icing limit. The scale has been set referenced to the chord length. Therefore, the average difference between the LEWICE prediction and the experimental data for lower surface icing limit is 6% of the chord.

It is very important to note that since these are absolute differences, one cannot state for certainty if this is a 6% over prediction or a 6% underprediction. A survey of the data in the spreadsheet shows that LEWICE **underpredicted** icing limit in approximately 2/3 of the cases and overpredicted the measured icing limit in 1/3 of the cases and that the average underprediction was greater than the average overprediction. The underprediction occurs because LEWICE does not in most cases capture the small scale ice feathers behind the main ice accretion.

This plot also shows the consistency of ice shape measurement since the experimental difference in icing limit is less than 2% of chord despite the fact that the icing limit consists of the small scale feathers which are considered difficult to trace with a pencil.

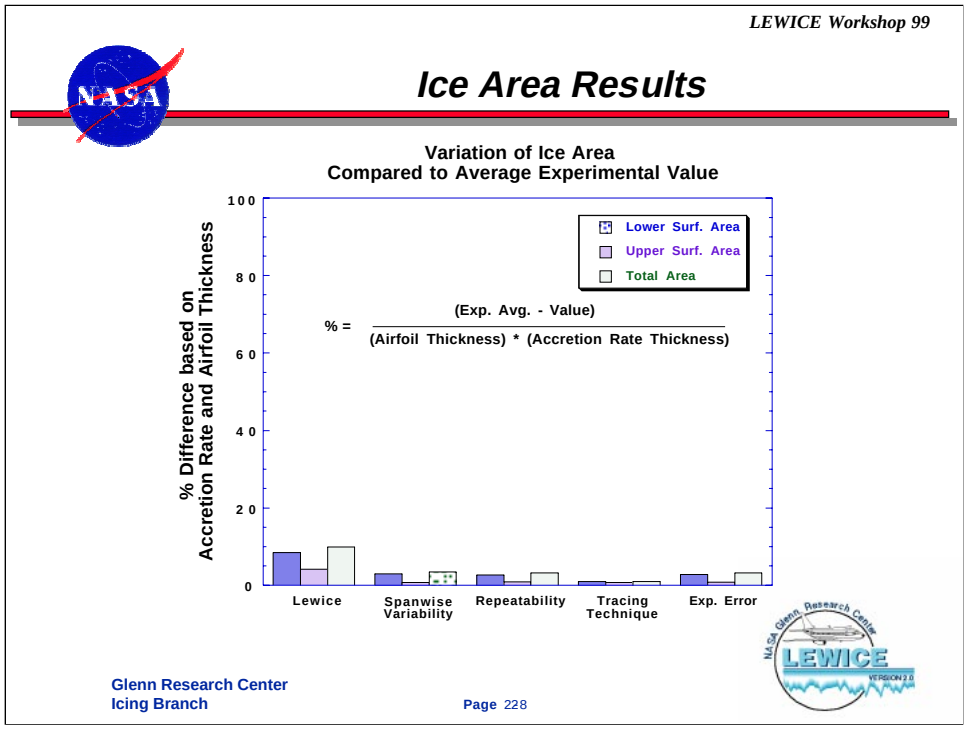


The accretion rate thickness is calculated from the equation:

Thickness = LWC * Velocity * Time / Ice Density

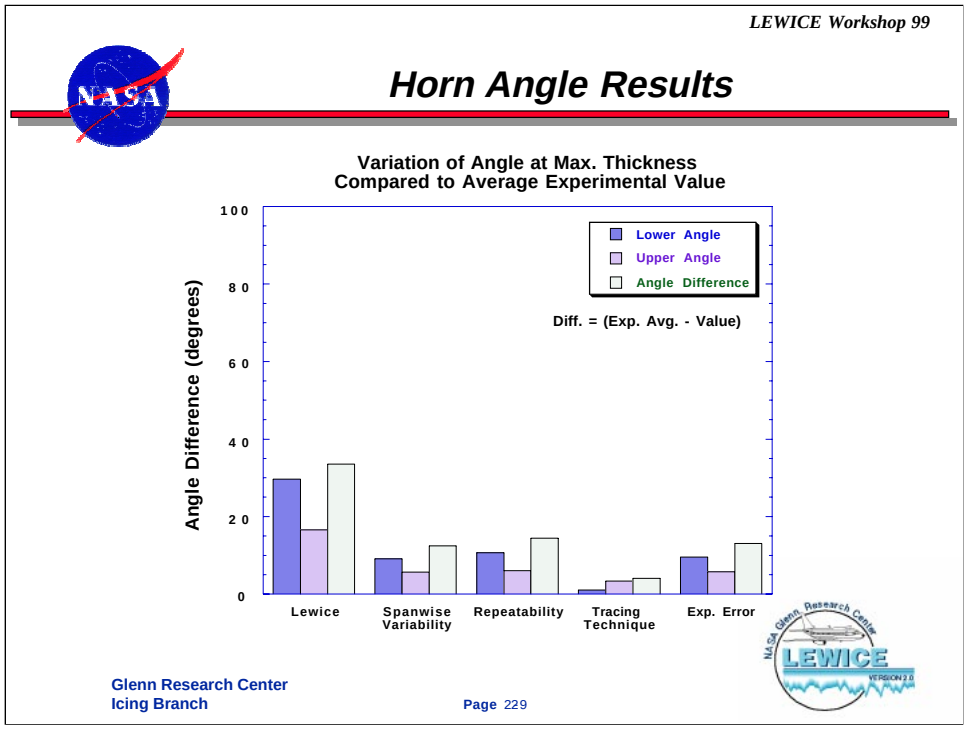
This was chosen so that cases with different accumulation rates could be compared fairly. This shows that LEWICE predicts the lower horn within 12% of experimental average value, the upper horn within 10% and the leading edge minimum within 6%.

To put these percentages into perspective, on a shape with a 3” maximum growth, the lower horn is predicted within 0.36” on average, the upper horn within 0.3” on average and the leading edge within 0.18” on average. On a smaller scale case, if the maximum ice growth is 0.3”, then each of the thickness differences decreases by a factor of 10 as well.

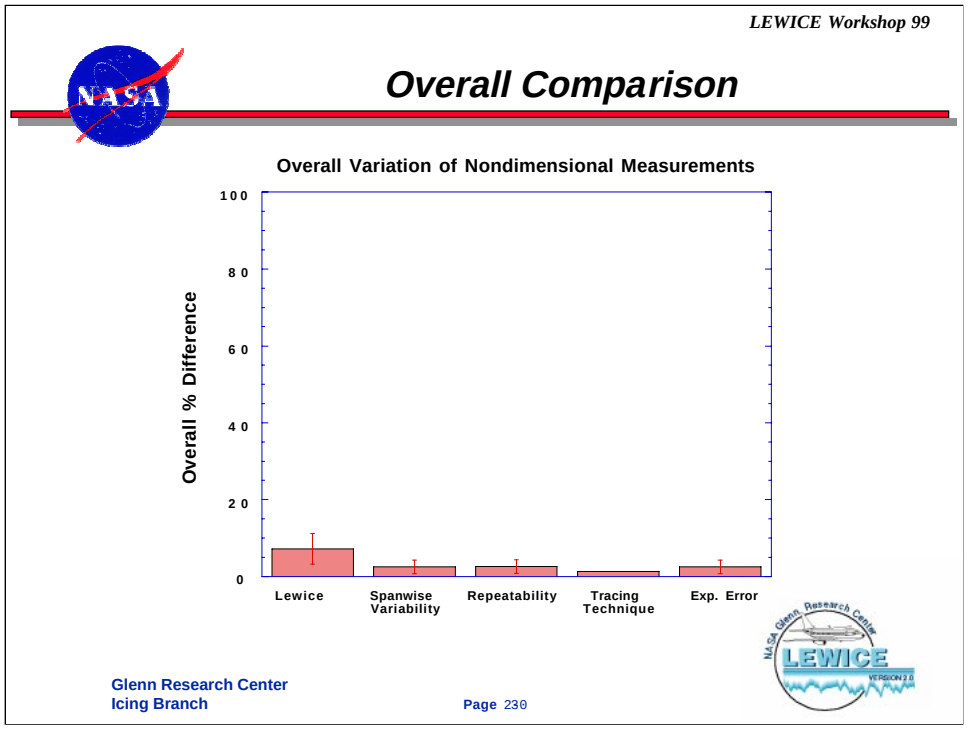


This plot shows the comparison of iced area for LEWICE and the experimental data. In this case, two scales are needed to non-dimensionalize the data. The first is a length scale, and the accretion rate thickness is a good index for this. The width scale was chosen to be the thickness of the airfoil.

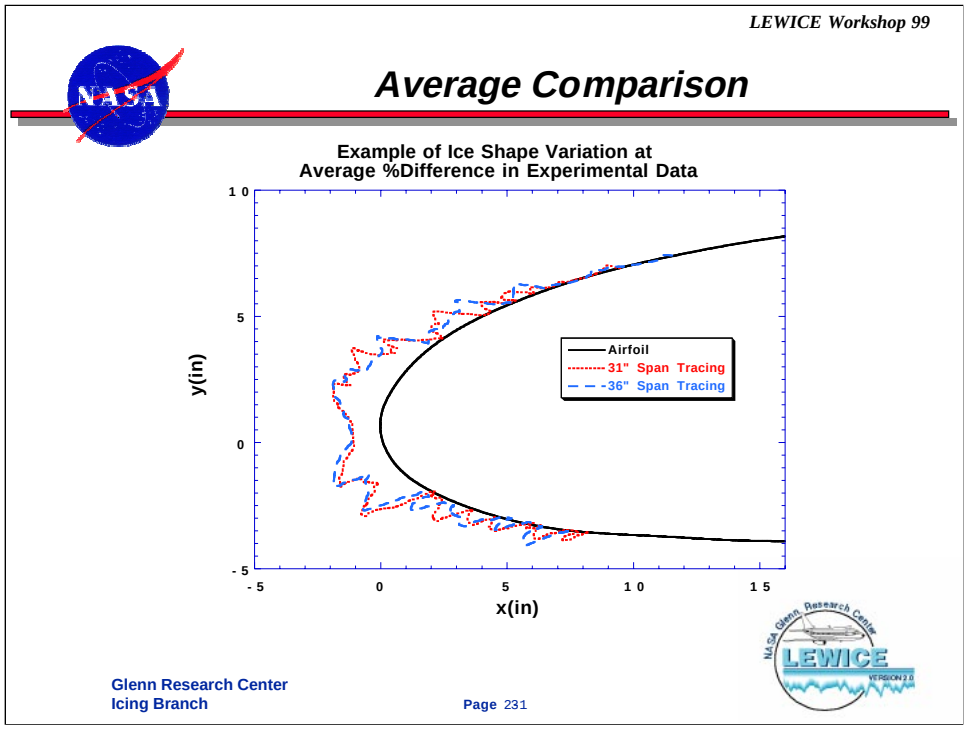
To put these measurements in perspective, for an experimental iced area of 1.06 inches (6 min accretion), the LEWICE area prediction was 1.3 inches which for that airfoil and that condition resulted in a 8.4% difference on the scale shown above.



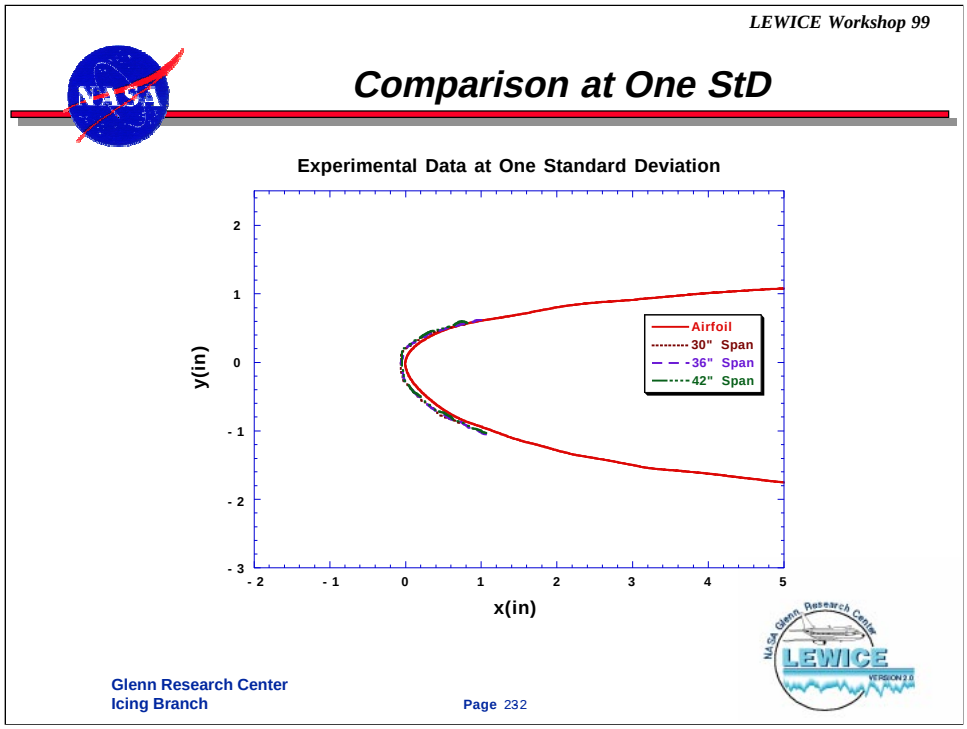
This plot shows the variation of horn angle for LEWICE and the experimental data. This data set does not need to be non-dimensionalized, so the results are presented in degrees. The angle difference is the angle made between the upper horn and the lower horn. This shows that LEWICE predicts the lower horn to within 30° on average and the upper horn to within 15° on average. The larger difference on the lower horn is due to the fact that on the experimental ice shape, there are usually several peaks and not a single lower horn as predicted by LEWICE. The LEWICE prediction is often between split peaks while not lying directly on any given shape. For an example, refer to Run 072504 from the GLC305 database which is used as benchmark case #2 in the LEWICE manual.



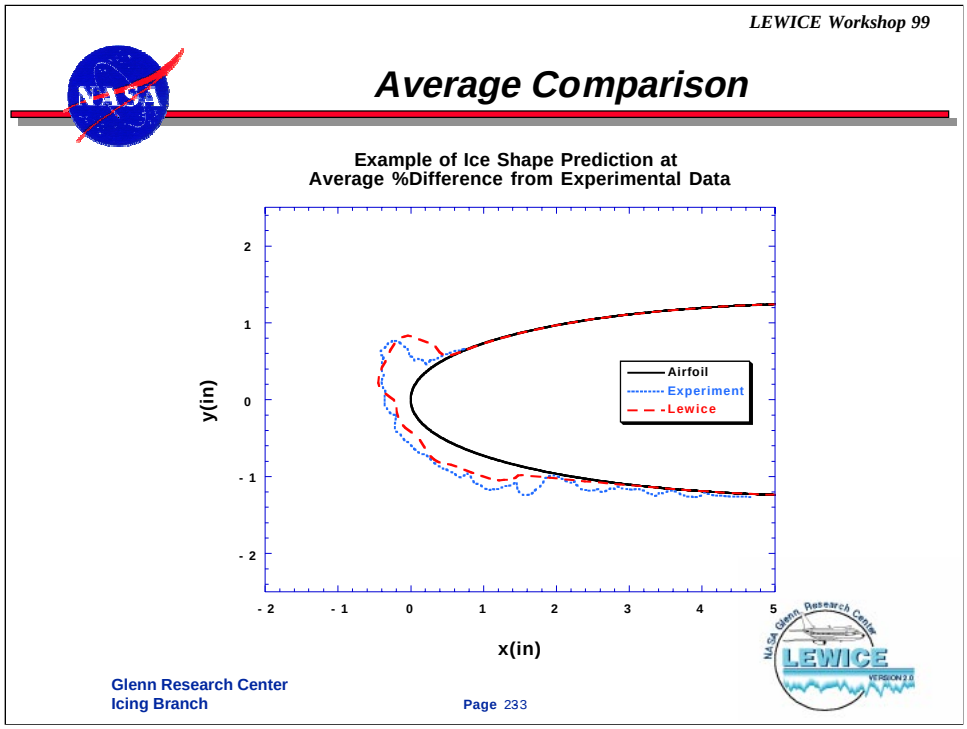
This factor takes the straight average of each of the previous 11 differences to arrive at the overall difference factor. To show some of the variability of the individual data, the statistical deviation of each result is also shown as an error bar on the plot.



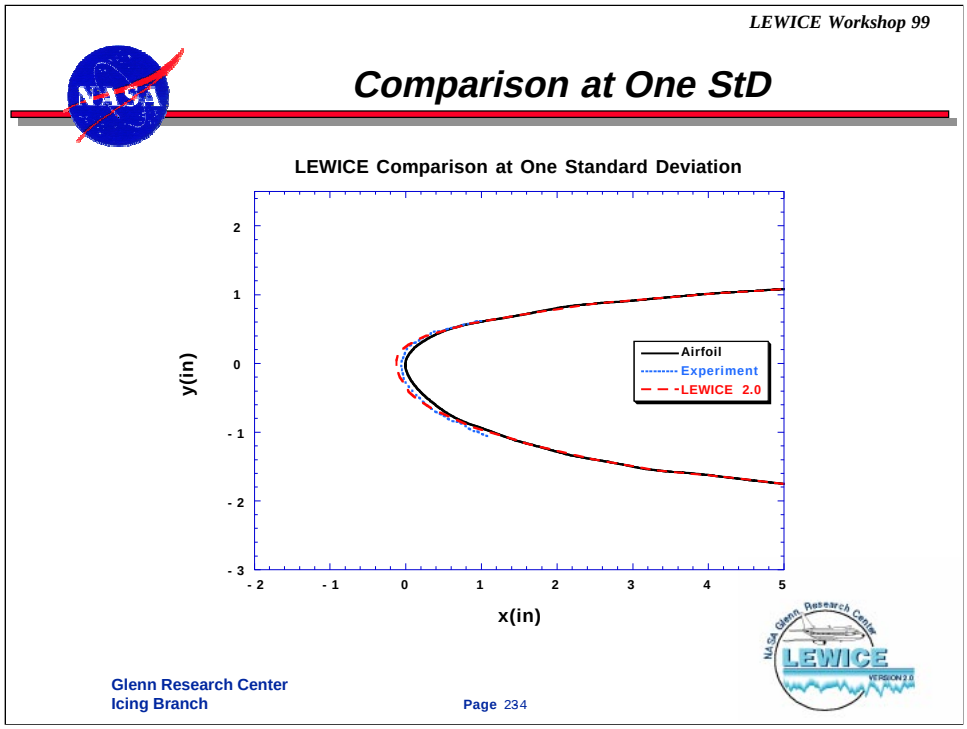
This plot was made to check if the quantitative assessment made agrees with the standard qualitative approach. In the above plot, the two experimental ice shapes differ from each other by 2.5% based on the overall difference factor. This represents the average variation of the experimental data. The specific case shown is a spanwise variation, although other exaples using repeatability could also be shown. In the qualitative sense, the ice shapes are very close to each other, so a low percentage such as 2.5% agrees with that assessment.



In this case, an example was chosen where the difference between the experimental ice shapes as at the average difference plus one standard deviation, or 4%. Since 4% is still quite small, there is still not much visible difference between the two shapes.



The previous plots showed examples of the variation in the experimental data. This plot shows the same qualitative comparison using the difference between LEWICE and the experimental data. The two ice shapes differ by 7.2% based on the overall difference factor. Indeed there is more difference between these two shapes that either of the two previous plots. Still, the comparison looks very good from the standard qualitative approach used previously. A 7.2% difference seems to fit both criteria nicely. (i.e. greater than the experimental variation, but a good prediction nonetheless)



This shows the difference between two ice shapes which are 11.5% different. This represents the LEWICE average comparison plus one standard deviation. By looking closely at the comparison, there is more ice at the leading edge in the LEWICE case. Since the total ice thickness is small, this leads to a larger percentage difference. This is typical of most cases where the percentage difference is higher than the average.

A qualitative review of the ice shapes predicted by LEWICE shows that less than 10% of the predictions can be considered “bad” in the qualitative sense. These cases are being investigated for insights into further improvements to the code.

Conclusions

- **Most extensive and comprehensive comparison to date**
- **Both experimental and computational ice shapes were quantified using 11 parameters**
- **Repeatability and Spanwise Uniformity variations very similar and very low**
  - **possibly due to natural chaotic variation of icing phenomena**
- **Tracing technique adds insignificant additional error**
- **Variation in non-dimensionalized experimental data is 2.5% ± 1.8%**
- **LEWICE prediction is within 7.2% ± 4% of the average experimental values**

The LEWICE 2.0 code has been designed to provide engineers with ice shape predictions which have a 90% confidence level in real time so the code can be used as an acceptable tool for certification and design of systems in icing. It exceeds both of these criteria by providing results which are on average 92.8% correct (7.2% difference from the experimental data) while running a 45 minute ice accretion simulation in under 6 minutes on a standard desktop PC.

Future Plans

- **Integrate thermal deice/anti-ice programs with version 2.0**
  - **Includes similar validation procedure**
    - » **Less data than for ice accretion validation**
- **Continue to improve integration with grid-based codes**
- **Droplet splashing**
  - **Large drop phenomena**
- **Continue thorough testing of subroutines**

This list represents our list of near-term research which is focused on LEWICE improvement. It is also planned to continue improvement of the physical models. Feedback is desired as to which models should be improved upon for future versions of LEWICE. Manpower resources are limited which may result in only the highest priority physical models being analyzed for future versions.

Discussion

- **Are results acceptable?**
  - **What part of predictions need improved?**
- **Is validation complete?**
- **What needs to be added?**
  - **Airfoils**
  - **Conditions**

If the user considers this validation effort to be incomplete, we would like to know what needs to be done (be as specific as possible).

Do you consider the errors too high?

For which conditions?

What other issues should be addressed for the code to be used in certification?

What should be our highest priorities in acquiring validation data?

Consider both which conditions and which class of geometries should be added.